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“MyanLitBot: An AI-Powered Literary Assistant for Myanmar Fiction”

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CS - 5112

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Outline

- Abstract
- Introduction
- Objectives
- Burmese Novel Corpus
- System Tools & Models
- System Evolution
- MyanLitBot’s Architecture
- System Workflow
- Feature Engineering
- Contributions & Model Implementation
- Evaluation
- Limitations & Future Work
- Conclusion
- References

Abstract

- The Burmese language suffers from a lack of large, structured digital corpora, unlike English, which benefits from resources such as the NLTK Brown Corpus and Gutenberg Corpus.
- This scarcity significantly limits academic research, educational applications and literary analysis in Burmese.
- This project introduces an intelligent system for exploring Burmese novels efficiently, enabling rapid retrieval of contextually relevant passages.
- By integrating semantic search with AI-driven literary interpretation, the system generates coherent, human-like analyses that remain faithful to the source text.
- This approach advances literary research, enhances educational resources and contributes to the preservation and accessibility of Burmese cultural heritage.

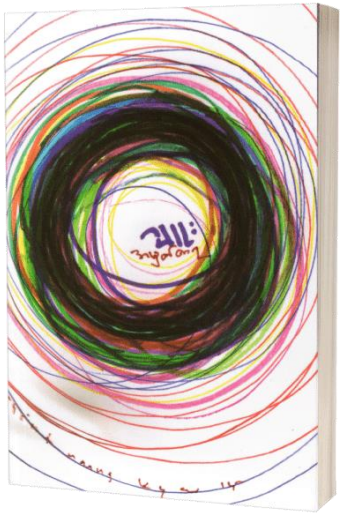
Introduction

- This project presents an AI-based system for explaining Burmese literary works.
- The system employs Retrieval-Augmented Generation (RAG) to combine retrieval and generation for accurate analysis.
- Generated explanations are carefully aligned with the content of the novels.
- The output emphasizes descriptive, academic-style literary interpretation.
- The system provides support for computational research in low-resource Burmese language processing.

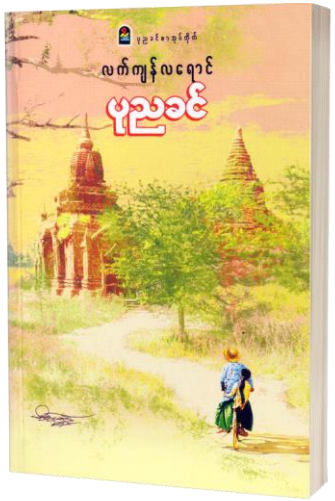
Objectives

- To develop an intuitive and robust platform for exploring and understanding Burmese literary works
- To enable precise retrieval of the most relevant story content, supporting detailed study and analysis
- To generate insightful, human-like literary interpretations that facilitate critical understanding of narratives, themes and characters
- To support education and literary analysis in the Burmese language
- To demonstrate the potential of AI technologies in overcoming the challenges of low-resource language processing

Burmese Novel Corpus

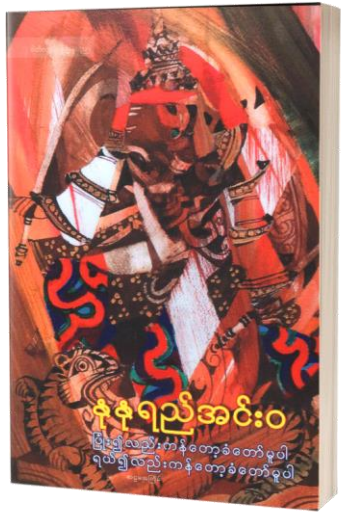


Genre: Love / Romance
Total Chapters: 8
Word Count: 13152
Character Count: 133186
Token Count: ~33297

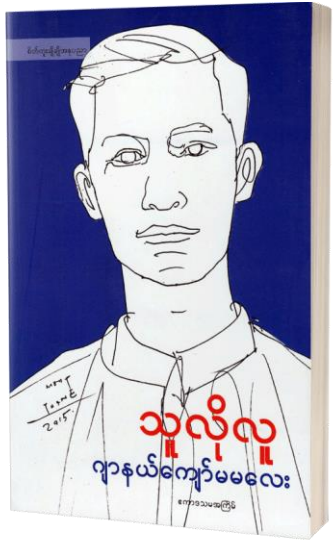


Genre: Love / Romance
Total Chapters: 21
Word Count: 42053
Character Count: 449925
Token Count: ~112481

Genre: Biography
Total Chapters: 42
Word Count: 37598
Character Count: 463129
Token Count: ~115782



Genre: Social Realism
Total Chapters: 14
Word Count: 18702
Character Count: 209449
Token Count: ~52362



Genre: Biography
Total Chapters: 34
Word Count: 76100
Character Count:790886
Token Count: ~197722

System Tools & Models

- **Python** – Core programming language (Version 3.10.12)
- **Sentence Transformers (LaBSE)** – Multilingual sentence embeddings (Version 2.2.0)
- **FAISS** – High-performance vector similarity search engine (Version 1.10.0)
- **Gemma-3-27B-IT**– Instruction-Tuned Large Language Model (27B parameters)
- **Scikit-Learn** – Analytical Utilities (Version 1.6.1)
- **Gradio** – Web-based user interface (Version 5.23.3)
- **Google Colab + GPU** – Model execution environment (Nov 2024 Release, Tesla T4)

System Evolution

Component	First Version	Second Version	Final Version
Novel Support	Single novel	4 novels	5 novels
Chunking	Basic	Sentence-aware	Optimized overlap
FAISS Index	IndexFlatL2	IndexIVFFlat	IndexIVFFlat + tuning
Retrieval	No reranking	Keyword reranking	Cosine similarity reranking
LLM Quality	Raw Output	Academic output	Literary analysis
User Interface	Textbox	Chat interface	Enhanced chat interface
Performance	Slow (40-45s)	Medium(30-35s)	Optimized(6-20s)

Why We Needed to Improve the System

From First Version to Second Version

Problems in First Version

- Duplicate sentences appeared in generated output
- Incomplete or abruptly ending explanations
- Limited scalability (only one novel)
- Retrieval sometimes returned weak or noisy context
- High LLM hallucination risk

Second Version Solutions

- Implemented clean_generated_text() function with sentence deduplication
- Added sentence completion check and removed imcompleted final sentences
- Implemented per-novel FAISS indexes supporting 4 novels simultaneously
- Added keyword-based reranking (rerank_chunks() function)
- Implemented strict prompt engineering with "ဝတ္ထုအတွင်း မတွေ့ပါ" fallback

Why We Needed to Improve the System (Cont'd)

From Second Version to Final Version

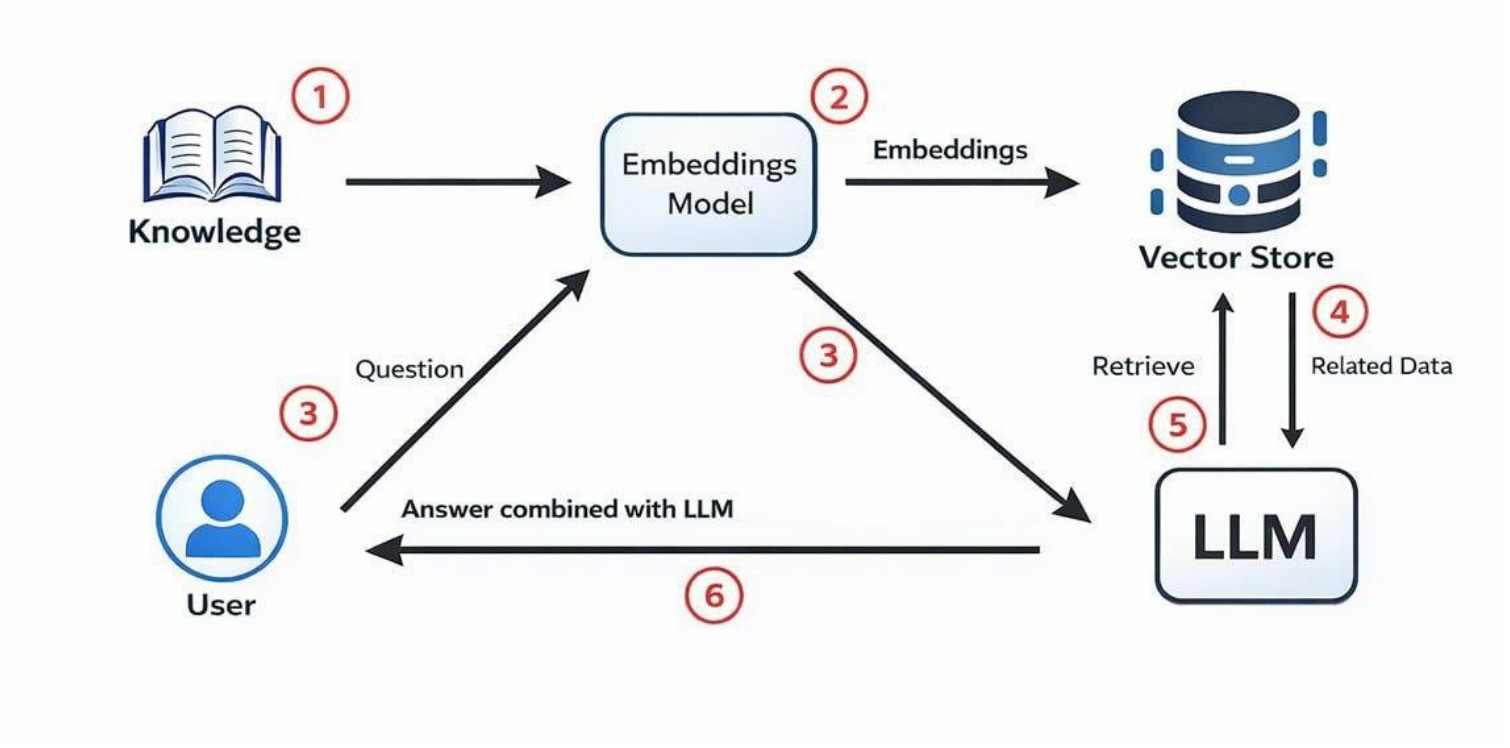
Problems in Second Version

- Out of Memory (OOM) Errors with Qwen2.5-7B (*7B parameter model requires ~14GB VRAM. Colab free T4 GPU only provides ~15GB VRAM, Model loading + embeddings exceeded memory limit*)
- Basic keyword reranking was inaccurate
- No response time tracking
- Indexes rebuilt every session
- Weak hallucination control

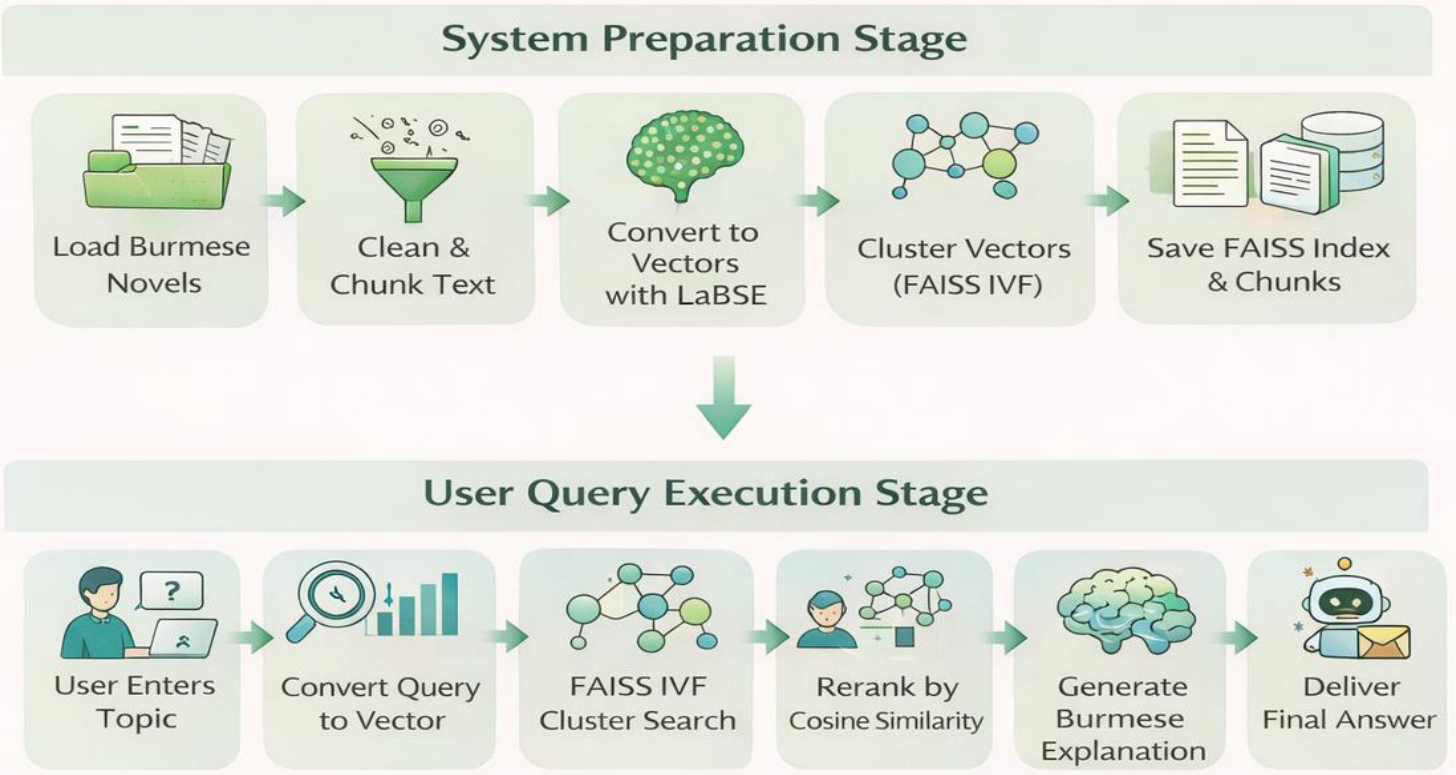
Final Version Solutions

- Switched to Gemma-3-27B-IT API (zero local GPU memory)
- Implemented cosine similarity reranking with LaBSE embeddings
- Added real-time duration measurement for each query
- Implemented persistent index storage system
- Enhanced prompt with strict semantic boundary rules

MyanLitBot’s Architecture



System Workflow



Data Preprocessing

- Load novel text from Google Drive.
- Remove extra line breaks.
- Strip leading and trailing spaces.
- Split text into sentences using punctuation (।, !, ?).
- Prepare text for chunking and embedding.

Feature Engineering

- Chunking
- Multi-Novel Isolated Indexing
- Sentence Embeddings and Semantic Retrieval
- Smart Semantic Clustering for Efficient Search
- Dynamic Cluster Adjustment
- Two-Stage Retrieval
- Hallucination Control & Human-Centric Prompts
- Output Deduplication and Cleaning

1. Chunking

What is Chunking?

Chunking is the process of breaking a long Burmese novel into **small, meaningful sentence-level text units** so that the system can **understand, search and retrieve context accurately**.

Why it is important?

- Preserves **complete Burmese sentence meaning.**
- Prevents **context loss** using sentence overlap.
- Improves **FAISS semantic search accuracy.**
- Produces **better LLM responses.**

How Chunking Works

Original Burmese Text

"သူမသည်မနက်ခင်းအလင်းရောင်ကိုကြိုဆိုနေသည်။…… ……..လေညင်းသံသည် စိတ်ကို အေးချမ်းစေသည်။သူ့ကို တွေ့သောအခါ နှလုံးသားက ထိတ်လန့်သွားသည်။……"

Sentence-Aware Chunks (max_chars = 500)

Chunk 1

"သူမသည် မနက်ခင်းအလင်းရောင်ကို ကြိုဆိုနေသည်။ ……..လေညင်းသံသည် စိတ်ကို အေးချမ်းစေသည်။"

Chunk 2 (with overlap one sentence)

"လေညင်းသံသည် စိတ်ကို အေးချမ်းစေသည်။သူ့ကိုတွေ့ သော အခါ နှလုံးသားက ထိတ်လန့်သွားသည်။……"

2. Multi-Novel Isolated Indexing

- Each novel is processed **independently.**
- Every novel has:
  - Its **own FAISS vector index (.index)**
  - Its **own sentence chunks file (.pkl)**
- Indexes are **saved to google drive** and reused.

Example: Multi-Novel Isolation
အမှတ်တရ: AMhatTaYa.index + AMhatTaYa.pkl
သူလိုလူ: ThuLoLu.index + ThuLoLu.pkl

Stored Files
.index → FAISS vectors (semantic space)
.pkl → Original Burmese sentence chunks

On System Restart

- Indexes are **loaded directly from google drive**
  - No re-chunking
  - No re-embedding

User Query:
"မောင့်ကို စိတ်ထိခိုက်စေသော စကားက ဘာလဲ"

- Search runs **only inside "အမှတ်တရ" index**
- No interference from other novels

3. Sentence Embeddings and Semantic Retrieval

- Each sentence becomes a vector.
  Example: "ဗိုလ်ချုပ်အောင်ဆန်းရဲ့အကြိုက်ဆုံးရုပ်ရှင်တွေပြောပါ“
  LaBSE converts it into a 768-dimensional vector.
  [0.12, -0.34, 0.87, ..., 0.05] ← 768 numbers
- Each number represents **a hidden feature of meaning** (e.g., emotion, action, context).
- Sentences with **similar meanings** have vectors **close together** in 768-dimensional space.

Example retrieved chunk:

> “…စစ်မဖြစ်မီက ရန်ကုန်တွင်ပြသွားသော (Wuthering Heights) (Rebecca) (Gone With the Wind) (Lady Hamilton) (Water Loo Bridge) (The Good Earth) (Suez) စသည်တို့မှာ ဗိုလ်ချုပ်နှစ်သက်သောသော ရုပ်ရှင်အချို့ပင်တည်း။…..”

Even if the words differ, LaBSE understands the **semantic essence**.

4. Smart Semantic Clustering for Efficient Search

- Groups similar meaning vectors into **clusters** first.

Cluster 1 : Love/Emotion

"ချစ်တယ်..."

"လွမ်းဆွတ်နေသည်..."

Cluster 2 : Nature/Setting

"မိုးရွာနေသည်..."

"ပန်းပွင့်များ..."

Cluster 3 : Memories

"မှတ်မိသေးသည်..."

"ယခင်ကတုန်းက..."

Cluster 4 : Daily Life

"ထမင်းစားသည်..."

"အိပ်ရာထချိန်..."

Before searching for relevant chunks, this system dynamically decides **how many clusters** to search based on the query complexity and corpus size.

Query: "အထီးကျန်ခြင်း...**"**

Step 1: Vectorize → [0.2, -0.4, 0.7, ...]

Step 2: Find nearest cluster → **Cluster 1** (Love/Emotion)

Step 3: Compare ONLY with vectors in **Cluster 1** (not all clusters)

Step 4: Return best matches from Cluster 1:

"လွမ်းဆွတ်နေသည်..."

5. Dynamic Cluster Adjustment

- nlist **(number of clusters)** is automatically calculated based on corpus size:
  nlist = max(10, min(100, len(embeddings)//5))
- Prevents **under-clustering** for small novels and **over-clustering** for large novels.
- Improves **indexing speed** and **retrieval accuracy**.
- Manual or capped clusters ensure **memory-efficient training**.

Example: Burmese Novels

အမှတ်တရ /A Mhat Ta Ya → 400 chunks → **80 clusters**

သူလိုလူ / Thu Lo Lu → 3000 chunks → **100 clusters (capped)**

Smaller novels → fewer clusters → maintain semantic coherence.

Larger novels → capped clusters → balance **speed, memory, and relevance**.

6. Two-Stage Retrieval (Search + Re-Ranking)

Stage 1: Fast Retrieval (FAISS IVF)

- User query → **vector embedding**
- **FAISS IVF cluster search** quickly retrieves relevant chunks.
- Handles **thousands of chunks efficiently** using pre-trained clusters.

Stage 2: Precision Re-Ranking (sklearn Cosine Similarity)

- Retrieved chunks → **encoded as vectors**
- **Cosine similarity** computed using sklearn.metrics.pairwise.cosine_similarit
- Chunks **ranked by semantic closeness** to the query.
- **Top 10 chunks selected** → ensures final result is most relevant.

7. Hallucination Control & Human-Centric Prompts

What is Hallucination?

Hallucination occurs when an AI model generates information that is not present in the source text, inventing characters, events, or details that do not exist in the original Burmese novel.

Why it is important?

- Prevents false literary analysis.
- Maintains academic integrity and trustworthiness.
- Ensures faithful representation of the author's work.
- Protects against misinformation about Burmese literature.

How Strict Hallucination Control Works

- LLM is **forbidden from using external knowledge**.
- Generation is **grounded only in retrieved novel passages**.
- If topic not found → returns a **controlled "Not Found" response**.

How Human-Centric Prompt Engineering Works

- Enforces **academic Burmese literary style**.
- Merges multiple fragments into **a single cohesive paragraph**.
- Disallows **chunk-level references** or mechanical phrasing.
- Eliminates **introductory and concluding filler text**.

8. Output Deduplication and Cleaning

- Duplicate sentences in the generated output are automatically removed.
- Repeated semantic content is filtered to avoid redundancy.
- Incomplete sentences are discarded.
- Abruptly ending sentences are removed.
- Ensures clean, stable, publication-quality explanations.

Example: Generated output before cleaning:
“ဗိုလ်ချုပ်အောင်ဆန်းသည် ရုပ်ရှင်များကို နှစ်သက်သည်။ ဗိုလ်ချုပ်အောင်ဆန်းသည် ရုပ်ရှင်များကို နှစ်သက်သည်။ စစ်မဖြစ်မီက ရန်ကုန်တွင် ပြသသော...”
Final output after cleaning:
“ဗိုလ်ချုပ်အောင်ဆန်းသည် ရုပ်ရှင်များကို နှစ်သက်သည်။”

Chunking Strategy

Aspect	First Version	Second Version	Final Version
Chunk unit	Character-based	Sentence-based	Sentence-based
Overlap	Character overlap	Sentence-level overlap	Sentence-level overlap
Sentence break risk	Possible	Eliminated	Fully Eliminated
Semantic quality	Medium	High	High

Retrieval Strategy

Aspect	First Version	Second Version	Final Version
Embedding Model	LaBSE	LaBSE	LaBSE
FAISS index Type	IndexFlatL2	IndexIVFFlat	IndexIVFFlat
Search Method	Linear Search	Cluster-based Search	Cluster-based Search
Reranking Method	None	None	Cosine Similarity
Retrieval Speed	Slow	Fast	Fast

Contributions & Model Implementation

- Implemented an **LLM-based Retrieval-Augmented Generation (RAG) system.**
- Designed **sentence-aware chunking** optimized for Burmese literary text.
- Used **LaBSE embeddings** for semantic representation of Myanmar language.
- Built **FAISS-based semantic retrieval** with IVF clustering for efficient context selection.
- Added **retrieval reranking** using cosine similarity to improve topic relevance.
- Applied **context grounding** with strict prompt rules to reduce hallucination.
- Supported **multi-novel corpus** with isolated indexes per novel.
- Integrated **output cleaning** to remove duplicate and incomplete sentences for academic quality.

Evaluation

Test Corpus

Novel Title (Burmese)	Novel Title (English)	Genre	Chunks
အမှတ်တရ(ဂျူး)	Memories	Romance/Drama	[491]
ဗိုလ်ချုပ်အောင်ဆန်း အတ္ထုပ္ပတ္တိ (သခင်ကိုယ်တော်မှူင်း)	General Aung San’s Biography	Biography	[1846]
ပြုံးရ၍လည်း ကန်တော့ခံတော်မူပါ ရယ်ရ၍လည်း ကန်တော့ခံတော်မူပါ (နုနုရည်အင်းဝ)	Smiling and Laughing in Reverence	Social Realism	[702]

Evaluation (Cont’d)

Test Questions

Question (Burmese)	English Translation	Category
ဇာတ်လိုက်၏ အဓိကစရိုက်လက္ခဏာက ဘာလဲ	What is the protagonist's main characteristic?	Character
ဤဝတ္ထုတွင် အဓိကပဋိပက္ခကဘာလဲ	What is the main conflict in this novel?	Conflict
အချစ်ကို ဘယ်လိုဖော်ပြသလဲ	How is love portrayed?	Theme
ဝတ္ထုပါ နေရာဒေသကို မည်သို့ပုံဖော်ထားသနည်း	How is the setting described?	Setting
စာရေးသူ၏ ရေးဟန်ထူးခြားချက်က ဘာလဲ	What is distinctive about the author's writing style?	Style

Evaluation (Cont'd)

Summary Observations Table

Aspect	Romance	Biography	Social Realism
Retrieval Quality	Excellent at finding emotionally charged dialogue revealing inner turmoil	Highly precise factual retrieval of historical events and character traits	Very good at finding socially charged dialogue capturing identity conflicts
Response Style	Warm, emotional language focusing on internal states	Formal, analytical tone examining historical significance	Dramatic, observational style analyzing social dynamics
Faithfulness to Text	Stays close to intimate scenes and emotional dialogue	Accurately reflects historical details and character traits	Captures essence of social dynamics through direct characterization

Evaluation (Cont'd)

Summary Observations Table

Aspect	Romance	Biography	Social Realism
Strengths	▪ Emotional depth ▪ Character intimacy	▪ Conflict analysis ▪ Psychological insight	▪ Factual accuracy ▪ Period details
Weaknesses/ Limitations	May over-focus on romance , missing other plot points	May miss peaceful , reflective moments	Less emotional range; can feel slightly distant

Findings

1. Genre Adaptation

The system successfully adapts to different literary genres.

Romance → Emotional and intimate tone

Biography → Analytical and factual style

Social Realism → Observational and socially focused interpretation

Adaptation is driven by genre-specific retrieved chunks.

2. Strongest Retrieval Performance

The Biography (ဗိုလ်ချုပ်အောင်ဆန်း အတ္ထုပ္ပတ္တိ) demonstrates the highest retrieval precision.Fact-based content (names, dates, events) aligns strongly with LaBSE embedding and cosine similarity search.

3. Most Challenging Genre

Romance (အမှတ်တရ) and Social Realism (ပြိုးရ်ှလည်းကန်တော့ခံတော်မူပါ ရယ်ရ်ှလည်းကန်တော့ခံတော်မူပါ) present subtle challenges.Risk: Over-focus on dominant emotional narrative when broader summaries are required.

4. Effectiveness of Cosine Similarity

Cosine similarity ranking works effectively across all genres.Top-ranked chunks consistently match query intent with high semantic relevance.

5. Unexpected Insight

The concept of “love” in the Biography was interpreted contextually (patriotism, admiration, friendship), not romantically.This demonstrates strong contextual adaptation across genres

Limitations & Future Work

Limitations

- The system currently supports a **limited number of Burmese novels**, restricted by available digitized text resources.
- Retrieval accuracy depends on the **quality of sentence chunking and embeddings**; subtle literary nuances may still be missed.
- **Low-resource language constraints** may cause minor retrieval errors due to limited pre-training data for Burmese models.
- The system focuses on **descriptive literary analysis** and does not yet support comparative or cross-novel reasoning.

Future Work

- Expand the corpus to include **more Burmese novels, short stories, and historical texts** across diverse genres.
- Introduce **cross-novel and comparative literary analysis** features.
- Develop **offline or lightweight deployment options** and a more accessible user interface.

Conclusion

- The project effectively addresses the critical gap in digital resources for Burmese literature, offering a practical and accessible solution.
- It provides a fast, accurate and context-aware system for exploring and analyzing novels, enhancing both academic and educational engagement.
- The integration of semantic retrieval and AI-based interpretation enables meaningful and reliable literary analysis.
- The project enhances access to Burmese literary resources, supporting research, education and cultural preservation.
- This work demonstrates the potential of artificial intelligence to advance low-resource language studies and establishes a foundation for future research and system expansion.

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